

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10	U04						*****VALVES*****			812	NIPPLE TO BE CUT FROM SS SCH XXS PIPE.
20										B1	ALL BOLTS ARE METRIC DIA.
20	U04						***** MODULAR VALVES *****			B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10
30											
30	U04		MODULAR DBB	K03AM10	0.5	2	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO - FULL SIZE TYPE 2BV	0	0		
40	U04	VAV	MODULAR DBB	K03AM10	3	3	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO - FULL SIZE TYPE 2BV	0	0	FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.
50	U04		MODULAR DBB	M03AM10	0.5	2	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	FS1	USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES.
60	U04	VAW	MODULAR DBB	M03EM10	0.75	1	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2	2		
70	U04	VAW	MODULAR DBB	M03EM10	1	1	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3	3	VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS
80										VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.
80	U04	VB2					***** BALL VALVES *****			VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.
90											
90	U04		BALL	A03AM09	3	3	BALL 300# RF INC 825 INC TRIM 625 FLOATING SOFT RB LO	0	0		
100	U04		BALL	A03AR11	4	4	BALL 300# RF LTCS + CLAD INC 825 INC 625 TRIM FLOATING SOFT RB LO	0	0		
110	U04		BALL	A03AR15	6	16	BALL 300# RF LTCS + CLAD INC 825 INC 625 TRIM TRUNNION SOFT RB GO	0	0	VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.
120	U04		BALL	B03AM09	0.75	2	BALL 300# RF INC 825 INC 625 TRIM FLOATING SOFT FB LO	0	0		
130	U04	VA5	BALL	B03AM09	3	3	BALL 300# RF INC 825 INC 625 TRIM FLOATING SOFT FB LO	0	0	VC1	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.
140	U04	VA5	BALL	B03AR11	4	4	BALL 300# RF LTCS + CLAD INC 825 INC TRIM 625 FLOATING SOFT FB LO	0	0		
150	U04	VA5	BALL	B03AR15	6	16	BALL 300# RF LTCS + CLAD INC 825 INC TRIM 625 TRUNNION SOFT FB GO	0	0		
160											
160	U04						***** BUTTERFLY VALVES *****				
170											
170	U04		BUTTERFLY	T03AR63	10	20	BUTTERFLY 300# RF LTCS + CLAD INC 825 INC TRIM 625 TOV METAL DBL FLG SHORT PAT GO	0	0		
180											
180	U04						***** GLOBE VALVES *****				
190											
190	U04		GLOBE	L03AM49	0.75	3	GLOBE 300# RF INC 825 INC 625 TRIM OS&Y HW	0	0		
200	U04		GLOBE	L03AR49	4	6	GLOBE 300# RF LTCS + CLAD INC 825 INC TRIM 625 OS&Y HW	0	0		
210	U04		GLOBE	L03AR44	8	10	GLOBE 300# RF LTCS + 825 CLAD INC 625 TRIM OS&Y GO	0	0		
220											
220	U04						***** CHECK VALVES *****				
230											
230	U04	VC1	CHECK	C03AM51	0.75	1.5	CHECK PISTON 300# RF INC 825 TRIM 625	0	0		
240	U04		CHECK	C03AM46	2	3	CHECK DUAL WAFER LUG 300# RF INC 825 INC TRIM 625 THROU BOLTED	0	0		
250	U04		CHECK	C03AR46	4	10	CHECK DUAL WAFER LUG 300# RF LTCS + CLAD INC 825 TRIM INC 625 THROU BOLTED	0	0		
260	U04		CHECK	C03AR53	12	16	CHECK DUAL DBL FLG 300# RF LTCS + CLAD INC 825 TRIM 625	0	0		
270											
270	U04						***** PIPE *****				
280											
280	U04		PIPE	5358511	0.75	1.5	PIPE SCH XXS SMLS LTCS A333 GR.6 PE	0	0		
290	U04	812	PIPE	5358511	0.5	0.5	PIPE SCH XXS SMLS LTCS A333 GR.6 PE	0	0		
300	U04		PIPE	5358513	2	2	PIPE SCH XXS SMLS LTCS A333 GR.6 BE	0	0		
310	U04		PIPE	5358515	3	4	PIPE SCH 160 SMLS LTCS A333 GR.6 BE	0	0		
320	U04		PIPE	5358514	6	10	PIPE SCH XS SMLS LTCS A333 GR.6 BE	0	0		
330	U04		PIPE	5358506	12	16	PIPE SCH 60 SMLS LTCS A333 GR.6 BE	0	0		
340	U04		PIPE	5358503	18	20	PIPE SCH 60 EFW LTCS A671 C660 CL.22 BE	0	0		
350											
350	U04						***** FITTINGS *****				
360											
360	U04		ELBOW	5392870	0.75	2	ELL 90 DEG LR XXS LTCS A420 WPL6	0	0		
370	U04		ELBOW	5392816	3	4	ELL BW 90 DEG LR SCH 160 LTCS A420 WPL6	0	0		
380	U04		ELBOW	5392600	6	10	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6	0	0		
390	U04		ELBOW	5392451	12	16	ELL BW 90 DEG LR SCH 60 LTCS A420 WPL6	0	0		
400	U04		ELBOW	5392876	0.75	2	ELL 45 DEG LR XXS LTCS A420 WPL6	0	0		
410	U04		ELBOW	5392822	3	4	ELL BW 45 DEG LR SCH 160 LTCS A420 WPL6	0	0		
420	U04		ELBOW	5392606	6	10	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6	0	0		
430	U04		ELBOW	5392498	12	16	ELL BW 45 DEG LR SCH 60 LTCS A420 WPL6	0	0		
440											
440	U04		TEE	5392879	0.75	2	TEE SCH XXS LTCS A420 WPL6	0	0		
450	U04		TEE	5392825	3	4	TEE BW SCH 160 LTCS A420 WPL6	0	0		
460	U04		TEE	5392609	6	10	TEE BW SCH XS LTCS A420 WPL6	0	0		
470	U04		TEE	5392501	12	16	TEE BW SCH 60 LTCS A420 WPL6	0	0		
480	U04		TEE RED	5392877	0.75	1.5	TEE RED SCH XXS LTCS A420 WPL6	1	2		
490	U04		TEE RED	5393073	1.5	2	TEE RED SCH 160/XXS LTCS A420 WPL6	3	4		
500	U04		TEE RED	5392821	3	3	TEE RED BW SCH 160 LTCS A420 WPL6	4	4		
510	U04		TEE RED	5392469	3	4	TEE RED BW SCH XS/160 LTCS A420 WPL6	6	10		
520	U04		TEE RED	5392661	6	8	TEE RED BW SCH XS LTCS A420 WPL6	8	10		
530	U04		TEE RED	5425179	6	10	TEE RED BW SCH 60/SCH XS LTCS A420 WPL6	12	16		
540	U04		TEE RED	5425170	12	14	TEE RED BW SCH 60 LTCS A420 WPL6	14	16		
550											

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - ALL SIZES SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 6 MM	SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 200 °C COMMENT: REFER NOTE 800	12/16/2024	BY OZAC	U04	Page 1 of 3	300K23

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
550	U04		CAP	5392880	0.75	2	CAP BW SCH XXS LTCS A420 WPL6	0	0	
560	U04		CAP	5392826	3	4	CAP BW SCH 160 LTCS A420 WPL6	0	0	
570	U04		CAP	5392610	6	10	CAP BW SCH XS LTCS A420 WPL6	0	0	
580	U04		CAP	5392502	12	16	CAP BW SCH 60 LTCS A420 WPL6	0	0	
590	U04		RED CONC	5392881	0.75	1.5	REDUCER CONC SCH XXS LTCS A420 WPL6	1	2	
600	U04		RED CONC	5395283	1.5	2	REDUCER CONC SCH 160/XXS LTCS A420 WPL6	3	4	
610	U04		RED CONC	5392827	3	3	REDUCER CONC SCH 160 LTCS A420 WPL6	4	4	
620	U04		RED CONC	5392462	3	4	REDUCER CONC SCH XS/160 LTCS A420 WPL6	6	10	
630	U04		RED CONC	5392458	6	8	REDUCER CONC SCH XS LTCS A420 WPL6	8	10	
640	U04		RED CONC	5392344	6	10	REDUCER CONC SCH 60/SCH XS LTCS A420 WPL6	12	16	
650	U04		RED CONC	5392341	12	14	REDUCER CONC SCH 60 LTCS A420 WPL6	14	16	
660	U04		RED ECC	5392882	0.75	1.5	REDUCER ECC SCH XXS LTCS A420 WPL6	1	2	
670	U04		RED ECC	5395286	1.5	2	REDUCER ECC SCH 160/XXS LTCS A420 WPL6	3	4	
680	U04		RED ECC	5392828	3	3	REDUCER ECC SCH 160 LTCS A420 WPL6	4	4	
690	U04		RED ECC	5392465	3	4	REDUCER ECC SCH XS/160 LTCS A420 WPL6	6	10	
700	U04		RED ECC	5392461	6	8	REDUCER ECC SCH XS LTCS A420 WPL6	8	10	
710	U04		RED ECC	5392645	6	10	REDUCER ECC SCH 60/XS LTCS A420 WPL6	12	16	
720	U04		RED ECC	5392646	12	14	REDUCER ECC SCH 60 LTCS A420 WPL6	14	16	
730	U04		OLETS	5504215	0.75	1.5	NIPOFLANGE 300# RF SCH XXS LTCS A350 LF2 CL.1	2	2	
740	U04		OLETS	5504214	0.75	1.5	NIPOFLANGE 300# RF SCH 160/XXS LTCS A350 LF2 CL.1	3	4	
750	U04		OLETS	5504216	0.75	2	NIPOFLANGE 300# RF SCH XS/XXS LTCS A350 LF2 CL.1	6	10	
760	U04		OLETS	5504219	0.75	2	NIPOFLANGE 300# RF SCH 60/XXS LTCS A350 LF2 CL.1	12	16	
770	U04						***** OLETS *****			
780	U04		OLETS	5505197	0.75	1	WELDOLET SCH XXS LTCS A350 LF2 CL.1	2	2	
790	U04		OLETS	5505396	0.75	1.5	WELDOLET SCH 160/SCH XXS LTCS A350 LF2 CL.1	3	4	
800	U04		OLETS	5505180	0.75	2	WELDOLET SCH XS/XXS LTCS A350 LF2 CL.1	6	10	
810	U04		OLETS	5505393	3	3	WELDOLET SCH XS/160 LTCS A350 LF2 CL.1	6	10	
820	U04		OLETS	5505182	0.75	2	WELDOLET SCH 60/XXS LTCS A350 LF2 CL.1	12	16	
830	U04		OLETS	5505183	3	4	WELDOLET SCH 60/160 LTCS A350 LF2 CL.1	12	16	
840	U04						***** FLANGES *****			
850	U04	FA8	FLANGE	5599703	0.5	0.5	FLG SCRDR (FNPT) REDUCING 300# RF LTCS A350 LF2 CL.1	0.75	1.5	
860	U04		FLANGE	5556481	0.75	2	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH XXS	0	0	
870	U04		FLANGE	5556480	3	4	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH 160	0	0	
880	U04		FLANGE	5556476	6	10	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
890	U04		FLANGE	5556452	12	20	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH 60	0	0	
900	U04	FS1	FLANGE	5556976	0.75	2	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH XXS	0	0	
910	U04	FS1	FLANGE	5557038	3	4	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH 160	0	0	
920	U04	FS1	FLANGE	5557036	6	10	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
930	U04	FS1	FLANGE	5556969	12	16	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH 60	0	0	
940	U04	FS1	FLANGE	5557145	3	4	FLG WN 900# RF LTCS A350 LF2 CL.1 SCH 160	0	0	
950	U04	FS1	FLANGE	5557141	6	10	FLG WN 900# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
960	U04	FS1	FLANGE	5557140	12	16	FLG WN 900# RF LTCS A350 LF2 CL.1 SCH 60	0	0	
970	U04	FS1	FLANGE	5557505	0.75	2	FLG WN 1500# RF LTCS A350 LF2 CL.1 SCH XXS BORE	0	0	
980	U04		BLIND	5615576	0.75	20	FLG BLIND 300# RF LTCS A350 LF2 CL.1	0	0	
990	U04						***** ORIFICE FLANGES *****			
1000	U04		ORIFICE FLG	5623950	2	2	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH XXS	0	0	
1010	U04		ORIFICE FLG	5623940	3	4	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH 160	0	0	
1020	U04		ORIFICE FLG	5623947	6	10	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
1030	U04		ORIFICE FLG	5623931	12	16	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH 60	0	0	
1040	U04						***** PIPE NIPPLES *****			
1050	U04	812					NIPPLE PBE - CUT FROM PIPE			
1060	U04	FS1					***** GASKETS *****			
1070	U04		GASKET	5673033	0.5	20	GASKET 300# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1080	U04	FS1	GASKET	5673103	0.75	16	GASKET 600# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1090	U04	FS1	GASKET	5673117	3	16	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1100	U04	FS1	GASKET	5672251	0.75	2	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1110	U04	B1					***** BOLTING *****			
1120	U04		BOLTING	5675770	0.5	99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
1130										

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - ALL SIZES SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 6 MM	SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 200 °C COMMENT: REFER NOTE 800	12/16/2024	BY OZAC	U04	Page 2 of 3	300K23

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1130	U04						***** BLINDS AND SPACERS *****			
1140										
1140	U04		SPEC BLIND	5749581	0.75	10	SPECTACLE BLIND 300# RF LTCS A516 GR.70	0	0	
1150										
1150	U04		PADDLE	5750100	12	20	PADDLE BLANK 300# RF LTCS A516 GR.70	0	0	
1160	U04		PADDLE	5750200	12	20	PADDLE SPACER 300# RF LTCS A516 GR.70	0	0	
1170										

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - ALL SIZES SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 6 MM	SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 200 °C COMMENT: REFER NOTE 800	12/16/2024	BY OZAC	U04	Page 3 of 3	300K23